Supporting Information

Enhancing anaerobic digestion of food waste by combining carriers and micro-aeration: performance and potential mechanisms

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Text S1 Food waste composition

The substrate used in this work is simulated FW to avoid the unnecessary effect of organic fractions variation on digestive efficiency. The simulated FW is composed by 400 g rice, 300 g cabbage, 200 g pork and 100 g tofu. These raw materials were firstly cooked in distilled water, and then were crushed to assure the particle size less than 5 mm. Finally, the crushed slurry was diluted to the volume of 2.0 L.

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Text S2 Measurement of Reactive Oxygen Species (ROS) Production

According to Limbach, Wick, Manser, Grass, Bruinink and Stark [1], the intracellular ROS production was determined using 2'7'-dichlorodihydrofluorescein diacetate (H2DCFDA; Molecular Probes, Beyotime), which can be oxidized to fluorescent dichlorofluorescein (DCF) by intracellular ROS. The digestates were centrifuged at 10,000 r/min for 10 min, then the pellets were washed using phosphate buffer (0.1 mol/L, pH 7.4) and resuspended in phosphate buffer with H2DCF-DA (50 μmol/L) and incubated in dark (35 ± 1 ℃). After 30 min, the pellets were centrifuged for 20 min and plated into a 96-well plate with phosphate buffer. Fluorescence measurements of DCF were conducted by a microplate reader (Synergy Lx, BioTek) with an excitation wavelength of 485 nm and an emission wavelength of 520 nm .

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Text S3 Electron Transfer System Activity

Electron-transporting system activity can be used to evaluate microorganism's respiratory activity and sludge bioactivity. In order to characterize the role of filler and micro-aeration on microbial activity in sludge during anaerobic fermentation, the ETS activity was determined, which was performed sing INT (2-(p-iodophenyl)-3-(p-nitrophenyl)-5-phenyltetrazolium chloride) methods with a small modification.Briefly, a 10 mL culture tube prepared for each sample was wrapped with aluminum foil to protect from light.Then, 2 mL of sludge sample anaerobic fermentation, 2 ml of Tris buffer (pH 8.4) and 0.5 ml of 0.2% INT solution were added into each culture tube, in sequence. After sealed and mixed evenly, the culture tube was incubated in an air bath shaker at 170 rpm and 37±1℃ for 30 min. 1 ml of 37% formaldehyde was subsequently added to terminate the ETS reaction, and the mixture was centrifuged at 10000 r/min for 5 min, follow by the supernatant was removed. Afterward, 5mL of methanol solution was added to extract the iodonitrotetrazolium formazan (INTF) by shaking at 170 rpm in the tube 37±1℃ air bath for 10 min. Finally, the supernatant obtained after centrifugation at 10000 r/min for 5 min was measured at the absorbance of 485 nm. The relative ETS activity was calculated by the following formula:

$$ETS\ \text{activity}\ (\%) = \frac{A_{485(\text{Experimental})}}{A_{485(\text{Control})}} \times 100 \qquad (1)$$

where A485 is the absorbance of the supernatant at 485 nm.

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Text S4 The method of EPS extraction

EPS can be divided into soluble EPS (SEPS), loosely bound EPS (LB-EPS) and tightly bound EPS (TB-EPS) [2], and the sum of all forms of EPS was the total EPS. PN and PS are the main components of EPS, accounting for 1-60% and 40-95% of the total composition of EPS, respectively [3]. Heat extraction method was modified to delaminate different EPS layers from WAS [4].The sludge suspension was first centrifuged for 10 min at 6000 rpm. The supernatant was collected as SEPS. Part of R-SEPS samples were stored and the rest were used for further extracting. The residual sludge in the tubes was then resuspended to the original volume with 0.05% NaCl, shaken horizontally at 150 rpm for 10 min under the temperature of 70 ^{o}C. The treated sludge was then centrifuged at 6000 rpm for 10 min to separate the solid and supernatant. The collected supernatant was regarded as the LB-EPS. The sludge mple (LB-EPS) left in the tubes was re-suspended in a 0.05% NaCl solution, then heated at 80 oC for 30 min, and finally centrifuged at 6000 rpm for 15 min to collect the TB-EPS.

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Text S5 Microscopy analysis

For scanning electron microscopy (SEM) analysis, cell slurries were fxed with 2.5% (v/v) glutaradehyde in phosphate-bufered saline, sequentially dehydrated with serial ethanol dilutions (20, 40, 60, 80, 95 and 100% (v/v) with every 10min per step). Te dried samples were coated with platinum and imaged using scanning electron microscope (SEM, Hitachi S-4800, Leica, Germany).

The presence of a carrier may create a physical barrier or provide an attachment point, thereby impeding or enhancing the contact of the colorant with the microorganisms. Therefore, in this study, sludge samples were collected for staining to assess the effectiveness of carriers in enhancing cell viability and optimizing microbial activity during anaerobic digestion after the introduction of micro-aeration. The cell viability of the sludge samples was stained using Live/Dead BacLight™ bacterial viability staining kit (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's instructions and observation using a confocal laser scanning microscope (CLSM) (ZEISS Axio Scan. Z1, Germany). Before the observations, the biofilm samples were rinsed with 50 mM PBS, then stained with a LIVE/ DEAD BacLight bacterial viability kit (Invitrogen, Carlsbad, CA, UnitedStates) and incubated in the dark for 15 min, and afterward rinsed with 50 mM PBS twice to remove the residual fluorochrome. The biofilm was measured under the z-stack mode. The green and red dyes of SYTO-9 and propidium iodide were used to stain the live and dead cells, which could emit green and dead fluorescence, respectively. The

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excitation and emission for SYTO-9 are 488 nm and 500–550 nm, for propidium iodide are 561 nm and 570–650 nm.

FISH Sample Preparation. The digested sludge samples collected from the reactor were immediately fixed in freshly prepared 4% paraformaldehyde solution. After fixation, they were washed with PBS buffer and then stored at -20℃ in a mixture of equal volumes of PBS buffer and 100% ethanol (maximum storage time: 6 months). The digested sludge samples were processed using the paraffin-embedded sectioning technique, resulting in sections with a thickness of 8μm. Oligonucleotide probes specific for bacteria (FITC-labeled EUB338mix probes) and archaea (Cy3-labeled ARC915 probe) were used in this study. Samples were examined using an Olympus BX41 (Olympus, USA) epifluorescence microscope. The green (bacteria) and red (archaea) fluorescence signals in the three sets of images were analyzed using the software Image J.

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Text S6 Microbial community and absolute quantitative PCR analyses

The changes in microbial community structure in different anaerobic digestion reactors were monitored by high-throughput sequencing of 16S rRNA genes. The total genomic DNA extraction from microbial communities was performed according to the instructions of the E.Z.N.A.® soil DNA kit (Omega Bio-tek, Norcross, GA, U.S.). The concentration and purity of DNA were detected by NanoDrop 2000 (Thermo Scientific, Wilmington, USA), and the quality of extracted genomic DNA was detected using 1 % agarose gel electrophoresis. PCR amplification of the V3-V4 hypervariable region of the 16S rRNA gene was performed with bacterial primers 338F (5'- GTGCCAGCMGCCGCGGTAA -3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3') using the thermocycling PCR instrument ABI GeneAmp® Model 9700 (ABI, CA, USA). The hypervariable region V3-V5 of the 16S rRNA gene was amplified with primer pairs Arch344F (5'-ACGGGGYGCAGCAGGCGCGA-3') and Arch915R (5'-GTGCTCCCCCGCCAATTCCT-3'). Then the DNA was purified and sequenced using the Miseq PE300 platform of Illumina. Operational taxonomic units (OTUs) clustering was performed on the sequences after quality control splicing according to 97 % similarity using UPARSE version 7.1, and chimeric sequences were identified and removed. The 16S rRNA database (eg. Silva v138) was taxonomically annotated for OTU using the RDP Classifier version 2.2, with a confidence threshold of 70%, and the community composition of each sample was counted at different species classification levels.

The DNA samples prepared in the previous step were reacted in 96-well plates using a fluorescence quantitative PCR instrument (model ABI7300, Applied Biosystems, USA), and the absolute content of 16s and ARCH archaeal genes in the

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samples was determined. The primers selected for the 16S rRNA gene were Eub338_Eub806, and those for the archaeal gene were Arch344F_Arch915R. The test results were analyzed using the Majorbio Cloud Platform (www.majorbio.com).

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Text S7 Detailed methods of untargeted LC-MS metabolomics

A certain mass of sample was accurately measured and ground, and the extract was added to the extract solution for metabolite extraction at low temperature, followed by centrifugation of the supernatant metabolite solvent for liquid chromatography mass spectrometry (LC-MS). UHPLC-MS/MS analyses were performed using a Vanquish UHPLC system (Thermo Fisher, Germany) and an Orbitrap Q Exactive TMHF-X mass spectrometer (Thermo Fisher, Germany) for UHPLC-MS/MS analysis.

Metabolites extraction

Accurately weigh 1000 ± 5 mg of sample into a 2 mL centrifuge tube and add a 6 mm diameter grinding bead,and the metabolites extracted using a 1000 μL methanol:water (4:1, v/v) solution.The frozen tissue was ground in a grinding machine at 50 Hz for 6 minutes at -10 ℃ and ultrasound at 40 kHz for 30 min at 5℃. The samples were placed at -20℃ for 30min to precipitate proteins.After centrifugation at 13000g at 4℃ for 15min, the supernatants were carefully transferred to sample vials for LC-MS analysis.

Chromatographic separation and mass spectrometer analysis

Chromatographic separation of the metabolites was performed on a Thermo UHPLC system equipped with an ACQUITY UPLC HSS T3 column (100 mm × 2.1 mm i.d., 1.8 μm; Waters, Milford, USA).Mobile phase A is 95 per cent water + 5 per cent acetonitrile (with 0.1 percent formic acid) and mobile phase B is 47.5 per cent acetonitrile + 47.5 per cent isopropanol + 5 per cent water (with 0.1 percent formic acid). The sample injection volume was 3 uL. The column temperature was maintained at 40℃. During the period of analysis, all these samples were stored at 4℃.

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The mass spectrometric data was collected using a Thermo UHPLC-Q Exactive HF-X Mass Spectrometer equipped with an electrospray ionization (ESI) source operating in either positive or negative ion mode.The samples were ionised by electrospray ionisation and the mass spectral signals were collected in positive and negative ion scanning modes, respectively. The data collected over the mass range m/z 70-1050 were acquired using data-dependent mass spectrometry (MS)/MS acquisition. The source parameters were as follows: spray voltage: 3500 V; capillary temperature: 325℃; heater temperature: 425℃.

Quality control sample setting

As a part of the system conditioning and quality control process, a pooled quality control sample (QC) was prepared by mixing equal volumes of all samples. The QC samples were disposed and tested in the same manner as the analytic samples. During instrumental analysis, a QC sample was inserted every 5-15 samples analysed to examine the stability of the entire assay.

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Table S1 Characteristics of FW and inoculated sludge

Characteristics	Feedstock	Inoculum
TS (g/L)	126.98± 0.60	64.60±0.44
VS (g/L)	123.07±0.41	22.11±0.50
VS/TS(%)	96.92±0.45	34.22±0.44
pH	4.90±0.12	7.49±0.20
TCOD (mg/L)	167215±4260	38045±2410
SCOD (mg/L)	51637±1390	1716±320
Crude fat(g/L)	19.96±0.50	NA
Crude protein(g/L)	18.36±1.70	NA
Carbohydrate(g/L)	71.40±4.00	NA

*Data shown as mean ± standard deviation of triplicate samples. NA: Not available.

The inoculum was settled naturally under anaerobic conditions for one day to remove the supernatant, and then the resulting sludge was filtered through a 20-mesh sieve to remove large impurities. The prepared inoculum was stored at anaerobic conditions of 35±1℃ for more than a week before use to eliminate the interference of residual biodegradable organic matter to subsequent AD process.

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Table S2 Details of the used polyurethane foam carrier and conductive carrier

Material	Shape	Size	Specific surface area (m^2/m)	Hanging membrane time (d)	Porosity (%)	Applicable pH	Carbon content
Polyurethane foam carrier	Cube	2cm×2cm ×2cm	> 4500	3-10	≥ 98%	6-10	/
Conductive carrier	Cube	2cm×2cm ×2cm	> 4500	6-10	≥ 95%	6-10	50-60%

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Table S3 Oligonucleotide probes used in this assay, target groups of each of them and hybridization conditions

Probes	Probe sequences (from 5' a 3')	Objective	Dyes	Formamida (%)	T (℃)	Time (h)
EUB338mix	5'-GCTGCCTCCC GTAGGAGT-3'	Bacteria	FITC	20	46	1.5
ARC915	5'-GTGCTCCCCC GCCAATTCCT-3'	Archaea	Cy3	35	46	1.5

*Data shown as mean ± standard deviation of triplicate samples. NA: Not available.

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Table S4 EPS content at the micro-aeration stage of the control (D0) reactor, polyurethane carrier (D20) reactor and conductive carrier (T20) reactor

Parameter	D0	D20	T20
S-EPS ($mg \cdot L^{-1}$)	290.69±5.91	403.96±4.86	364.58±5.05
LB-EPS ($mg \cdot L^{-1}$)	135.36±4.02	106.44±7.43	91.77±9.14
TB-EPS ($mg \cdot L^{-1}$)	216.04±5.15	586.25±9.54	689.50±11.96
TEPS ($mg \cdot L^{-1}$)	642.09±4.72	1096.65±6.24	1145.85±8.75
PS ($mg \cdot L^{-1}$)	175.25±5.17	187.43±5.57	205.99±3.93
PN ($mg \cdot L^{-1}$)	466.84±3.90	909.22±4.33	939.85±8.66

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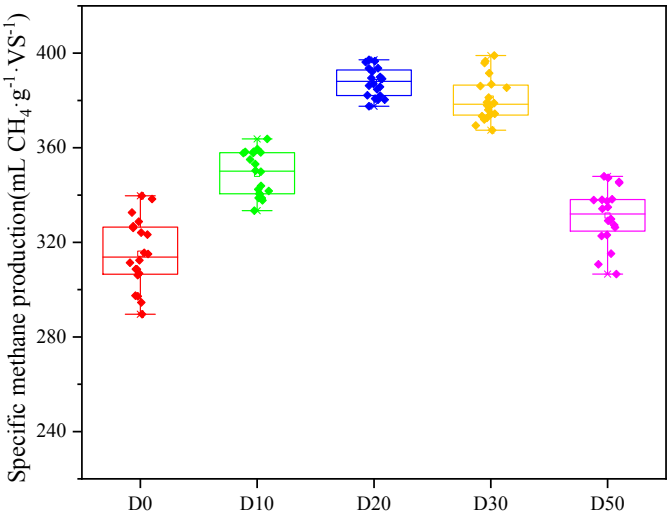


Figure.S1 The specific methane yield at different filling ratios in the reactor

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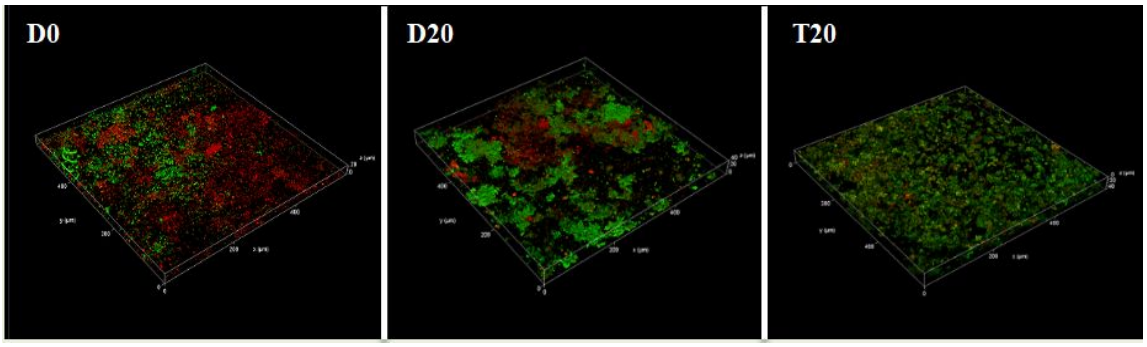


Figure. S2 Live/dead bacteria images of sludge collected from D0, D20 and T20 in in stage Ⅲ

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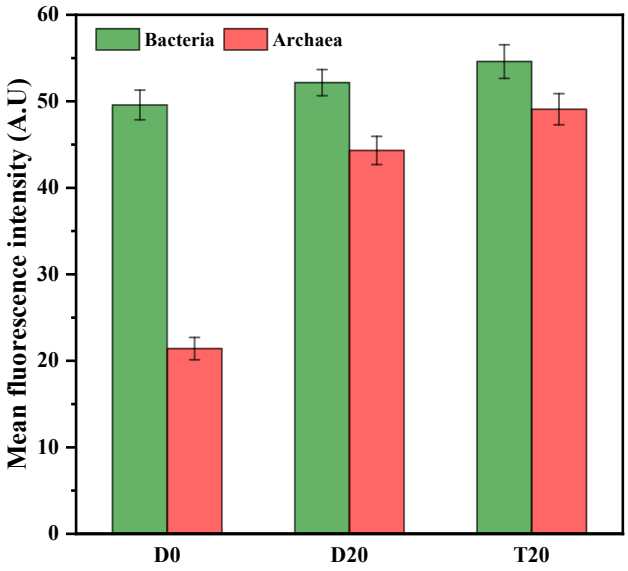


Figure S3 The analysis results of green (bacteria) and red (archaea) fluorescence signals in the two sets of images using the software Image J.

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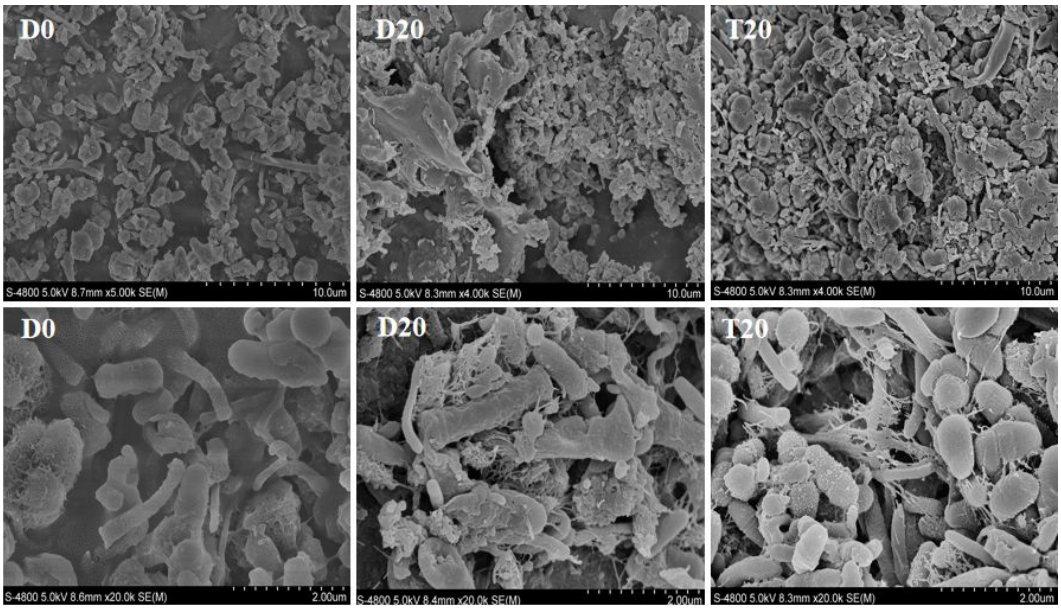


Figure S4 Morphological characteristics of digested sludge (D0) and filler surface (D20 and T20) at10um and at 2um

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Figure S5 Polyurethane carrier (above) and conductive carrier (below)

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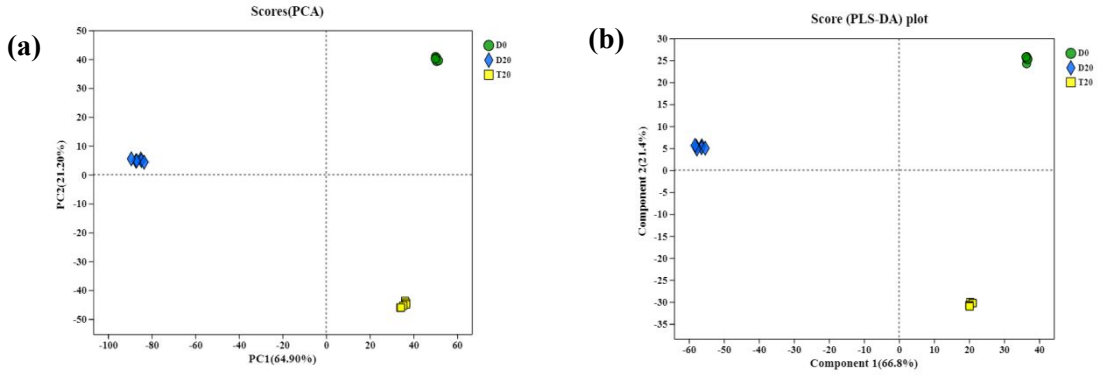


Figure S6 Statistical analysis of metabolites extracted from D0Ⅲ, D20Ⅲ and T20Ⅲ samples (a) principal component analysis (PCA) and (b) partial least-squares discriminate analysis (PLS-DA) score plots of metabolic profiles.

The results of the principal component analysis (PCA) and partial least squares discriminat analysis (PLS-DA) showed that the distinction between different groups is significant, indicating that their metabolism are significantly different (Figure. S7a and b). Metabolites in AD process are mostly derived from materials produced by microbial activity and mediators formed by organics decomposition, with metabolites from both sources being powered by microbial activity [5]. Therefore, the results showed that the carrier combined with micro-aeration changed the metabolic pathways within the microorganisms.

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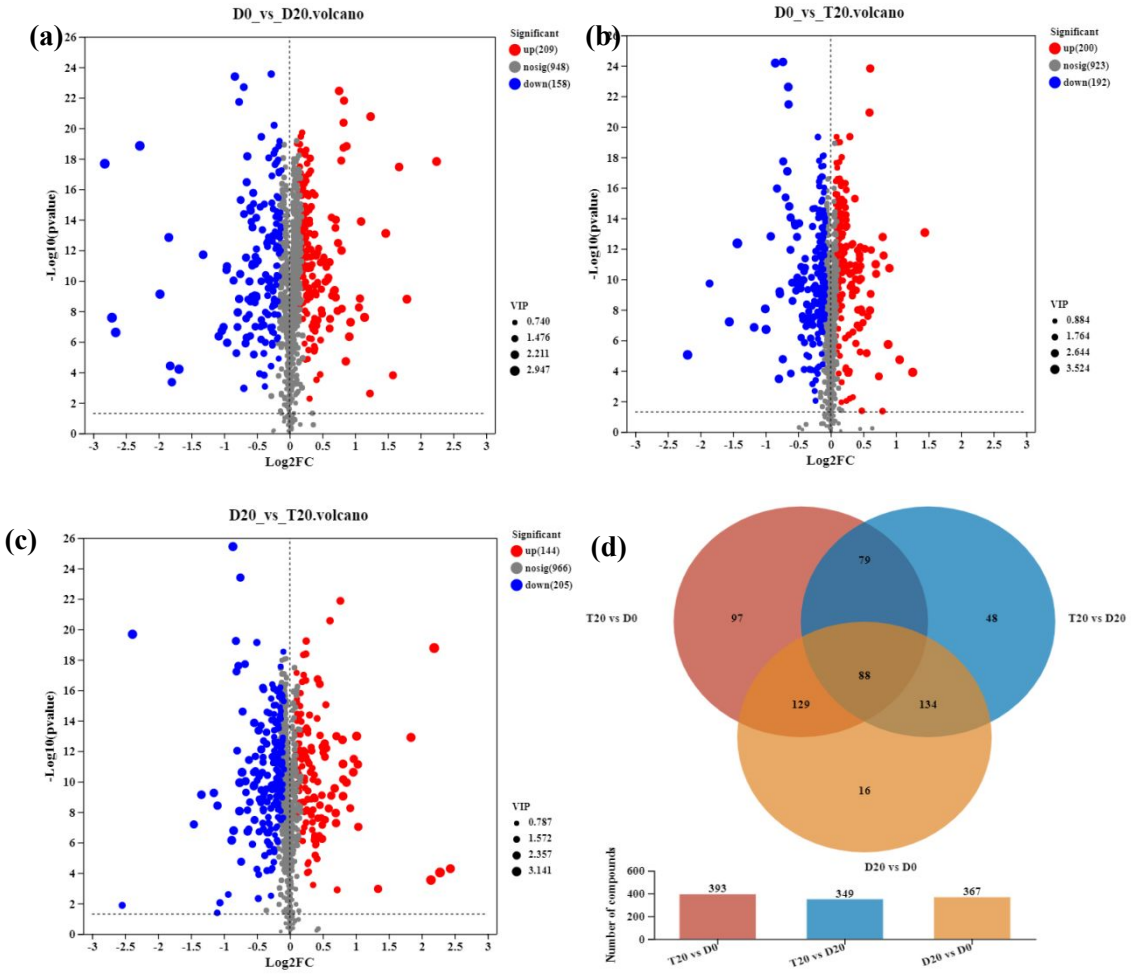


Figure.S7 Volcano plot (a. D0 vs D20; b. D0 vs T20; c. D20 vs T20) and Venn diagram (d) of differential metabolites in stage III

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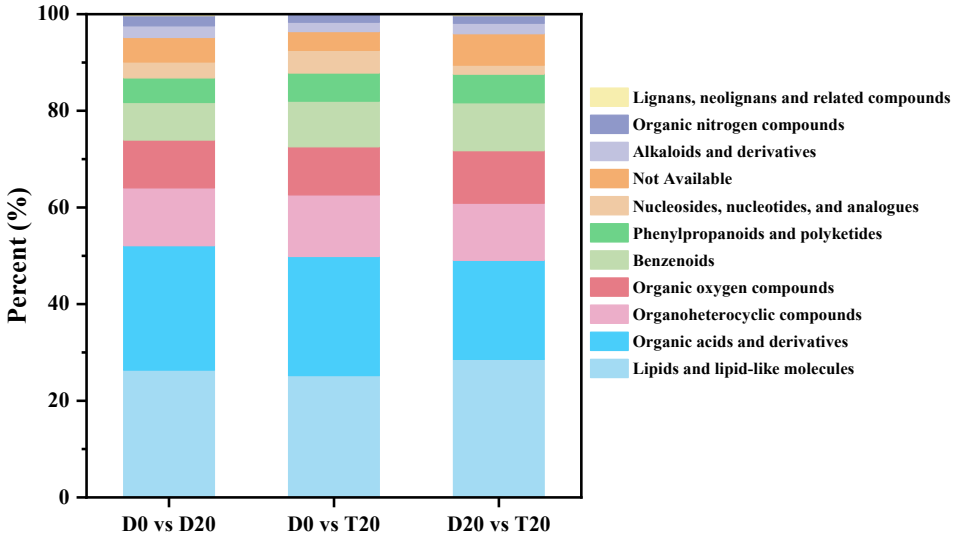


Figure S8 The classification of the differential metabolites at the "superclass" level of the HMDB database

The groups of differential metabolites were visualized and analyzed as volcano plots (Figureure S8a-c), and significant differential metabolites were screened as 393 (T20 vs D0), 349 (T20 vs D20) and 367 (D20 vs D0), respectively. Studies have shown that the molecular and metabolic interactions between T20 and D0 were the key factors affecting the conversion of organic matter and the production of methanogenic metabolites during micro-aeration. The lowest number of differential metabolites was observed in the T20 group versus the D20 group, which may be related to the presence of the carrier. Moreover, it was also observed that these down-regulated metabolites were more numerous than the up-regulated metabolites (Figure S8, supporting statement). Venn diagram was used to collect significant differential metabolites between different digestive groups (Figure S8d). The results showed that there were 161 non-duplicated differential metabolites and 88 metabolites of differential metabolites were present in all the different control groups. The classification of differential metabolites in the three reactors based on the superclass

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level in the Human Metabolome Database (HMDB) showed that most of them belonged to "Lipids and lipid-like molecules", "Organic acids and derivatives", "Organic heterocyclic compounds" and "Organic oxygen compounds" (Figure. S9).

(1) Limbach, L.; Wick, P.; Manser, P.; Grass, R.; Bruinink, A.; Stark, W. Exposure of engineered nanoparticles to human lung epithelial cells: Influence of chemical composition and catalytic activity on oxidative stress. **2007**, *41* (11), 4158-4163.

(2) Yu, G. H.; He, P. J.; Shao, L. M.; Zhu, Y. S. Extracellular proteins, polysaccharides and enzymes impact on sludge aerobic digestion after ultrasonic pretreatment. *Water Res* **2008**, *42* (8-9), 1925-1934.

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(4) Dai, Y.; Huang, S.; Liang, J.; Zhang, S.; Sun, S.; Tang, B.; Xu, Q. Role of organic compounds from different EPS fractions and their effect on sludge dewaterability by combining anaerobically mesophilic digestion pre-treatment and Fenton's reagent/lime. *Chemical Engineering Journal* **2017**, *321*, 123-138.

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